NAME: _____ Geometry (FL B.E.S.T.)

6.1 Pythagorean Theorem

In a right triangle with legs a and b and hypotenuse c, the Pythagorean Theorem says $a^2 + b^2 = c^2$. The converse lets you classify any triangle as acute, right, or obtuse by comparing $a^2 + b^2$ with c^2 (where c is the longest side). This worksheet goes beyond one-step triples: expect radicals, multi-step applications, and classification with careful ordering of sides.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Legs 11 and 60. Find the hypotenuse.

2. Hypotenuse 14, one leg 6. Find the other leg in simplest radical form.

3. Sides 12, 35, 37. Is the triangle right? Show the test.

4. Sides 6, 9, 11. Acute, right, or obtuse?

5. A 29 ft ladder's base is 20 ft from a wall. How high does it reach?

6. TV screen 27 in by 36 in (rectangular). Find the diagonal.

7. Sides 13, 14, 15. Classify.

8. Legs 5 and $5\sqrt{3}$. Find the hypotenuse.

9. Hypotenuse $2\sqrt{13}$, leg 4. Find the other leg.

10. A baseball diamond is a square with side 90 ft. Find the distance from home to second (exact form).

11. Sides 5, 12, 14. Right triangle? Justify.

12. Right triangle: one leg is 8, hypotenuse is 3 more than the other leg. Find all three sides.

13. A rectangular park is 40 m by 42 m. You walk the two short sides and one long side, or you cut the diagonal then walk one short side. Which route is shorter, and by how much?

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14. An isosceles triangle has sides 10, 10, and base 12. Classify the apex angle (between the equal sides). Hint: compare $10^2 + 10^2$ with 12^2.

15. A wire from the top of a 15 m pole to the ground is 25 m long. How far from the base is the anchor? If the wire slips so the anchor moves 8 m farther from the base, how high on the pole does the wire now reach?

✓ Answer Key

1. 61
2. $4\sqrt{10}$
3. Yes; $12^2 + 35^2 = 37^2$
4. Obtuse ($36 + 81 = 117 < 121$)
5. 21 ft
6. 45 in
7. Acute ($169 + 196 = 365 > 225$)
8. 10
9. 6
10. $90\sqrt{2}$ ft
11. No; $25 + 144 = 169 \neq 196$ (obtuse)
12. Legs 8 and 15; hypotenuse 17
13. Diagonal+short is shorter by 14 m
14. Acute ($100 + 100 = 200 > 144$)
15. First distance 20 m; new height 7 m